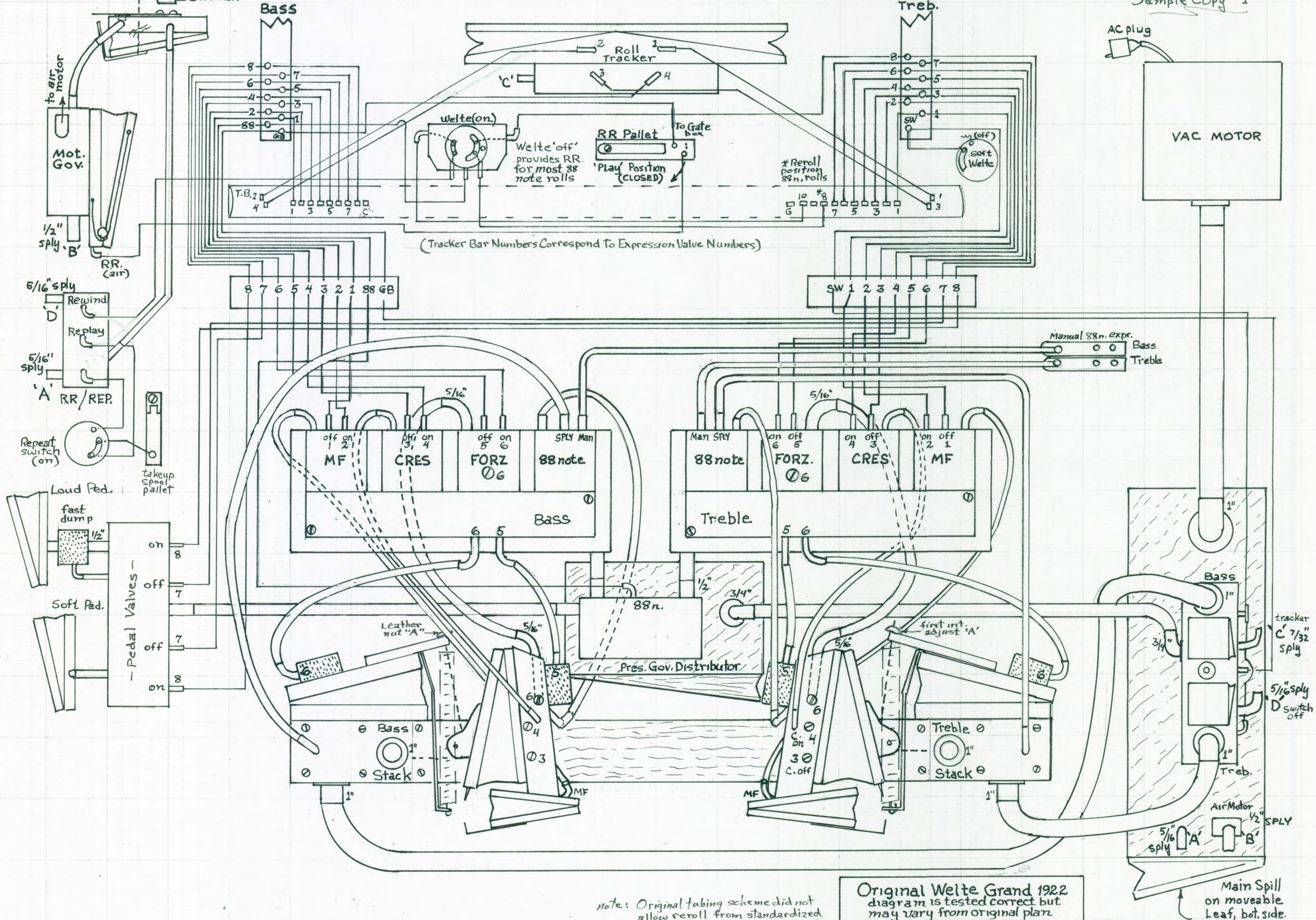
Tubing Chart For Original Welte Mignon Grand

297 E. 133rd st. NYC.

Sample Copy #1



note: Original tubing scheme did not allow reroll from standardized rolls incl. Ampico or original roll labels.

Original Welte Grand 1922
diagram is tested correct but may vary from original plan
8/2011 Craig Brougher

Regulating the Welte Licensee and Original Welte-Mignon

By Craig Brougher

Although the individual components can be tested for leaks and go-no-go checks on the bench, the Welte Licensee and Original Welte Mignon depends upon precise roll speed for the final expression timing checks and would be difficult and impractical to be roughly pre-adjusted on a bench setup like other reproducer systems.

If regulating a grand, most of the initial adjustments can be performed while the grand is resting on its side, on a grand skid board. This gives easy access to all components. Only the Welte allows this type of adjustment because all of its valves are of the guided metal stem type, and do not need to be vertical in order to work at low intensities. In the Welte system, exact tempo is required to play its expression correctly.

In the Original Welte grand the stack is hidden and difficult to adjust. The piano action itself will be the Staib-Aberschind which is designed for the stack to nest behind it and contact its whippens, rather than the ends of its keys. Since these whippens rest at slightly different heights and attitudes, an adjustment gauge needs to be made to transfer each whippen's height to the stack where the finger's capstan can be adjusted accordingly. It is also sometimes necessary to add stack risers (shims) inside the piano shelf which houses the stack. There are two rods crossing the stack's path which must be removed each time the stack is inserted, so leave them out until all static adjustments are finished. These rods are the sustain and hammer rail actuators. They will cause you to break striker fingers if they are left in while reinserting the stack. There may also be tips of trap mounting screws protruding through the shelf and should be turned down.

Removing that type stack after the player is tubed up it's very easy to do since the stack has a single pouch nipple strip which is removable by 13—1"x #6 wood screws. Raise the strip out of the way and tie it and its tubing high, unscrew the 4 flathead screws holding the stack to the shelf bottom, remove any other hoses in the way, and pull it right out. It is the easiest stack to remove there is.

Initial Procedure

a. Disconnect the air motor drive chain so the test roll can be operated by hand. Spool up the test roll, switch the power on, bypass the first tests and go to the Welte note test if desired. At this point, you can set stack pressure with nut 'A.'

b. With WELTE transforming switch "**on**," and the modulator switch "**normal**," test all the notes for operation. You can do this with the piano action out of the original mignon grand and simply watch the strikers, or you can leave this test for later, because dynamics at the keyboard are not the initial setup consideration. To make the notes play louder, adjust leather nuts "A" which we'll call first intensity. These are on the ***expression regulator*** arms. By connecting a vacuum gauge to each bass and treble end of the stack when this is done, a pianissimo level can be initially guessed at. Try 5" H_2O, initially.

Preliminary Explanation to Expression Tests

The expression tests on the Welte test roll are crucial to its operation because they are mainly timing tests. With such a system it is virtually impossible to adjust the piano strictly by listening to its music, as can be done by other systems.

The MF (mezzo-forte) hook check is performed with various roll tests. However compromises must be made so that each adjustment balances with the others which come later. The original manuals do mention the need to compromise but they fail to acknowledge the fact that some compromises are more crucial than others and some claims are unnecessary. Keep in mind that it is necessary to work back and forth and to keep rechecking the timing, and that there is no such thing as absolutely perfect adjustments. Stress the final performance. That's what is important.

Also, when one adjustment is made it affects the others. The reason is because the Welte works strictly by timing the actuation and de-actuation speed of a single pneumatic each in the bass and treble. That means the speed at which a vacuum closes a pneumatic is screw-adjustable, and that vacuum is working against an adjustable spill which is continually dumping that vacuum out again. So when you are adjusting closure speed by just using the crescendo or forzando adjustment, if the #3 screw (decrescendo) is too far out, your adjustments for both will be completely off and everything will have to be repeated, no end. Our method keeps rechecks to a minimum.

Different models of Welte did not even place the expression screws in the same relative position with each other, so you cannot number them according to any drawing available. You have to know by rebuilding it which screw does what, and then label that screw adjustment. The decrescendo is just a vacuum leak to air with a screw throttle interposed. It is called "cresc. off" or #3, or "O." <u>*The forzando adjustment on the Original Mignon*</u> is a large diameter set screw (#6) in series with the forzando valve travel screw so only one is necessary. Start the adjustments with the valve depth screw open for 1/16" valve travel, and plan to adjust only with the set screw for the 6-pump strobing test #10.

Setting Pump Pressure

a. Remove pump spill muffler block if pump is a bellows pump run by a motor. If it is a Welte suction box, disregard.
b. With vacuum gauge connected in place of a note tube on the side you are about to test, set the “pianissimo” (first intensity) to 5” with the leather nut A, and then close down the expression pneumatic by hand. Gauge should read about 35” of vacuum for a grand, 32” for an upright. If not, adjust the spill valve. Repeat test with other half of stack and expression. If 35”(32”) isn’t reached, tighten spill valve spring.
c. If player is Original Welte-Mignon with a suction box, put a “T” in the 7/32” reservoir vacuum line going to the top action, and connect a gauge. Turn the spill pallet crank CCW on the large reservoir to raise the pressure. Suction boxes do not achieve as high a pressure as a positive displacement bellows pump, but have more latitude by providing more volume of pressure. With no notes playing, watch the vacuum raise until turning the spill crank does not increase pressure further. Stop turning, reverse crank direction until you see the needle budge backward again. That will be the spill setting.
d. Test the operation of the **soft and sustain pedals**. Watch the height that each one raises and set for optimum. This may be done first by pulling the respective #8 tube and checking mechanical operation first, then cancelling off with #7 and rechecking piano action operation from the trackerbar holes to make sure the lift is adequate. Hammers should raise ¾” to perhaps 7/8”. Dampers to just clear.
e. The **forzando** works best when set faster than called for. So when you reach test #10, consider that as long as the first two strobes are outside the MF hook arc, then where the others drop isn’t important. I’ll cover this more in detail later.

Test Procedure (with trackerbar covered, first)

(<u>Skip to test 4</u>). Tag all the tubes to be removed so you don’t get them mixed up. It is assumed that each test mentioned will also be repeated in the bass, and so will not be repeated over and over again, here. Test 4 on the roll is really the most important one.

If someone has covered the MF hook with leather, clean it off, first. Scrape off all glue or shellac and polish it up with steel wool. (*They may “click” together when the blade is captured*.) You will see later why covering this hook with leather is entirely incorrect.

Repeatability is all-important in this type of Reproducer. And repeating the tests over and over again is likewise a necessity, as well as going back through the sequence of tests to the beginning and rechecking them after a week of “playing in.”

An accurate, high quality vacuum gauge (0-40, preferably) such as a Magnahelix, which is about the best available and reasonably priced, is very necessary for regulating the Welte.

1. Set the first intensity on both stack halves to about 5" by adjusting leather nut "A."Then close each expression pneumatic by hand and release. If the 1st int. doesn't return to the set 5" tighten the expression valve's return spring a bit. It should return to within $\pm$ ½"H_2O. If close, then the following adjustments may take care of the problem, so continue anyway.

2. Compare treble and bass maximum pressures when the expression pneumatics are closed. If they are not equal, there may be a leak between the regulator and the stack, or the large regulator spring on that side is weak, or the other side is too strong. Usually the fixed spring hook is adjustable and allows them to be equalized.

3. Correct position of the MF hook. **Do not use the Welte method** of setting this hook. It is totally wrong. They also call both blade and hook, "*a hook*." Very ambiguous. **This adjustment must be set with a vacuum gauge**.

When the blade blocked by the MF hook, the pressure should read <u>about</u> 60% of the maximum stack pressure. So if the pump pressure reads 35," the MF level should read 20." 32" would have a MF of 19," and 25" would have a MF of 15." Multiply the pump pressure by .6 to get MF.

Allow the blade to be captured by the hook now and the new reading should read no greater than 2" higher than the MF reading. If the angle of engagement creates more than 1-1/2 or 2" H_2O difference, add felt to the MF pneumatic cover to prevent it from closing so deeply. That can decrease the difference, as long as the hook hasn't been bent. It must be a right angle hook. *This measurement necessitates why the blade cannot be padded*.

All music is set up around the desired MF intensity. If for some reason you want a different intensity, now is the time to set it up. Decrease the MF intensity and you compress **the soft intensities** in resolution. Increase the MF intensity, you compress the **loud intensities**. These are just principles of operation, and they cannot be ignored.

The most important element of expression is the degree of soft gradations up to MF. That means when forzando is called for it's always going to be a turkey shoot to hit an accented note during a rapidly closing expression pneumatic on the fly. It will NEVER be exactly in the same place twice, anyway. That's because of the variants of mass and velocity product (*which change depending on the percentage of closure*) and all the other pressure fluctuations, even though the expressions utilize their own regulated air. When the pump is at capacity the spill cannot compensate so variations due to instantaneous volume of accented notes jar the entire system. Only the large reservoir springs have energy stored, but a reservoir reacts and cannot *anticipate*, much less in milliseconds, so it's of little help. Nailing the exact displacement of a flying forzando able to close in .1 second each time is impossible, so loud accents are approximate.

4. Remove and mark tubes 1, 2, 3, and 4 from the trackerbar, at the junction blocks. Add stoppers so that you can operate their expression valves, one at a time. With a digital watch, making sure the blade is not *captured* under the hook, **time the *decrescendo* of the expression pneu. first**. The blade should go from touching the MF hook to fully open pneumatic again in **2.0 sec**. That will change with other adjustments to follow, but for the moment, adjust screw #3 (**cresc. off**) to get changes in the speed of return. Repeat again and again. To get more accurate results, use a pair of spring-loaded pallet valve switches on #3 &4. We'll do actuation later. (*The 'Licensee' & 'Original' manuals are just backwards. Ignore them.*)

   Sometimes **cresc. off** allows you to spot a small vacuum leak to the expression pneumatic even though it's *working*, and that requires troubleshooting the valves. It can be caused by either **cresc.** and/or **forz.** valve not seating perfectly. Too much leak, in turn, can be caused by a tight cancel pouch buoying up the stem above the lock valve, or a bad seal around the inside valve seat.

   Practically speaking, both valves must leak just a tiny bit, anyway. So if the pneumatic they are powering were perfectly airtight, it would slowly close off. The #3 screw decrescendo dumps for both of them constantly, even when it is not used. That means the #3 screw has to have a minimum opening just to take care of unwanted bleeding. It is one reason to have small bled vents to air in the crescendo valve chamber. However, these are not always sufficient. The actual problem is in the difficulty of balancing one or more input bleeds against a single exhaust bleed, which must also take into consideration the tiny vacuum leakages unpredicted and inconstant.

   If the expression pneumatic isn't returning to within ± 1/2" of the originally set first intensity and leakage vacuum is not the problem, adjust its compass spring. Too weak and it will crescendo too quickly. Too strong and it will never close fully. One reason for poor repetition is the leak mentioned above. In the *Original* Welte-Mignon and all Weltes, there is a separate vacuum regulator (*of the throttle type*) providing separate pressure for just the expressions alone. **The expression valves with their expression regulator (with the large extension spring) are separate from stack pressure**. This is the principle of all Weltes, of course.

Many old “diddled-with“ Weltes have stretched-out large extension springs. *These should no longer be used* and can be replaced by springs of approximately similar gauge, diameter, and length which are pre-stretched to your *approximate specifications*. Roughly, the spring on one *Original Welte-Mignon* was 6.5” long x 1” OD, and tested (*when I got it*) .7-.8 lbs/first inch (I suspect too light). Then it pulled 6.5lbs/second inch. But a 6.5” L x ¾” OD works well. It can be found at hardware stores and prestretched to pull 6.8 lb/second inch. That seemed to work very well. *The important thing is that the spring is a linear constant within travel limits*. Unless the 2 springs are equal you’ll have a problem, but many original springs seem to be stretched out, so I go by the second inch, and then restretch the new springs to this. (*Use a digital gauge designed to weigh luggage.*) That first inch stretch difference is adjusted out with the spring hook.

5. Adjust the speed of crescendo next using “P” screw, or cres.#4. This time you will be uncorking #4 tube and starting your stopwatch simultaneously. It works better to kink a drop tube, pull its stopper, then let the kink go or hit a pallet button as the stopwatch is started. Check this multiple times, too. The crescendo ***actuation timing*** to the MF hook is **2.5 sec**. Watch the vacuum gauge and as soon as the needle reaches MF pressure, the watch must read **2.5**, or very closely.

Test with Roll

1. **Roll test #2**. First intensity test only so that the notes of test #3 can be heard. This has already been done by now.
2. **Roll test #3**. Note test. Welte “on,” Volume lever “normal.”
3. **Roll test #4**. This has already been done.

Forzando is Italian for fast crescendo (6) and fast decrescendo (5). It means “forcing. “ The forzando valve gap is adjustable by a depth screw on its top and is the only adjustment on the Licensee. On the *Original Welte Mignon*, there is also a series set screw. With both, I’d suggest to set the valve at about 1/16” travel and then use the set screw. The Original Welte Mignon manual is wrong regarding their own “2, 4, 6 test” #4. It is impossible to lock the blade under the hook when the hook actuates first. Ignore it. The writer obviously never regulated a Welte in his life. (*be aware of other mistakes including small errors in the Licensee schematic. A poor job of documentation plus duplicitous procedures. It requires art, understanding, and a good ear to set a Welte.*)

4. **Roll test #5**. First forzando test. 4, 6, and then 3, 5 repeated 4 times. Forzando and crescendo on at the same time, and then both off. The pair stays on for .1 second, at which time the blade should reach the hook "*but not quite touch*." The second pair of 3, 5 turns off crescendo and the expression pneumatic should instantly return. This is repeated 4 times, yet they don't actuate the hook to tell (?)

   This is really only generally correct because *any measurement with forzando actuation speed cannot be predicted*, so when they say should not touch the hook, that can mean anywhere in its free operating space. I set it so that it comes within 1/16" to 1/8" of the hook, and I manually close the hook so I can tell.

5. **Roll test #6**. Same as test #5 except the expression pneumatic passes the hook position by about ¼" (or so). For your amusement only.

6. **Roll test #7**. Same as test #6, except stack pressure should go to maximum, and will fully close the expression pneumatic. This one makes sense. Adjust #6 screw.

7. **Roll test #8**. For your entertainment mainly. What Welte calls "*locking the expression pneumatic*" is incorrect, again. If the MF hook arrives just ahead of the pneumatic blade, then it doesn't "lock" the blade. It "STOPS" the blade. To lock means to capture, so it blocks further movement. That's all it does. I suspect this regulation booklet was only checked for spelling and syntax by Miss Grammar.

8. **Roll test #9**. This test just makes sure the MF hook can still beat the blade when forzando #6 is added. If not, weaken the MF pneumatic spring.

9. **Roll test #10**. The idea of this test is to check the roll speed at 80 by "strobing" the expression pneumatic with the #6 intensity at a rate which will cause it to progress toward the MF hook. There will be 6 pulses. On the 6th pulse, the blade should be supposedly even with the MF hook. The 7th perforation will be the #5 fast decrescendo which should return the expression pneumatic instantly. Any test with forzando is a speed test and relies on the #3 (or "O") decrescendo screw adjustment set earlier for sufficient dropout between strobes. As long as the first two pulses are outside of the MF hook's arc, then it just doesn't matter what the remainder do.

In order to adjust the progression speed, change the forzando valve screw on the valve block or the forzando set screw #6 on the *Original*. If you want to experiment, then open the set screw fully and start closing the forzando valve screw until you see a definite speed change of the expression pneumatic closure. At that point, open the valve screw a little and close the set screw until you are controlling the expression pneumatic with the set screw, instead. Now you have only one screw to adjust instead of two.

10. **Roll tests #11, 12**. Provides a moment of relaxation for the technician. These tests are worthless and prove absolutely nothing, but you might enjoy them. Besides, when the piano is playing a roll, the Welte is the most entertaining of all the reproducers to crawl under. *I think the expression pneumatic will crack pecans*.

11. **Roll test #13**. The object of this test is to see if the MF hook will ever bind up on the blade and be unable to return. There are three reasons that could happen. First if someone has covered the hook with leather, and second, if someone has bent the hook to some angle other than 90 degrees. Third, if the MF pneumatic compass spring has been weakened.

12. **Roll test #14**. The hammer rail test. Part 1: We want to measure the hammer rail travel, so on grands, drop a scale through the treble strings onto the top of a hammer and measure the distance of the hammer at rest, then actuated. Depending on the piano action of course, the hammer rail should travel the hammer between 3/4" and 7/8." This can be a static test without the roll (8B).

    Part 2: The quickness of the hammer rail to get into place before a set of notes are struck. Take out lost motion and shorten the travel to accomplish this. A stop screw on the soft pedal pneumatic is also a very good idea here.

13. **Roll test #15**. Damper pedal test (8T). Check for smooth, quiet operation and a quick operation test as well. One thing the roll won't do is allow you to test damper lift height. Make this a static test first, and then when all the dampers just clear the strings and no more than that, let the roll test their quickness.

14. **Roll test #16**. This is a test of the loud pedal when playing 88 note rolls. This test doesn't apply to all Welte Mignons. Some were not equipped to play 88 not sustain. Switch Welte switch to Welte "off" or "88 note, " and loud pedal switch

to “on.” Any problem will be found in valve block “L.” Follow line “U” through the Loud Pedal switch, transforming (Welte) switch, back to 3B on the trackerbar.

15. **Roll tests #17 & #18**. These are reroll tests.#17 tests an 88 note roll to rewind, while #18 tests a Welte roll to rewind. The original Welte Mignon was not rigged to recognize 88 note reroll, either. However, using my schematic, you are able to automatically rewind most 88 note rolls. The 10T Welte reroll hole in my schematic is able to switch with the 2T for 88 note rewind.

This completes the regulation of the **Welte Licensee** and **Original Welte Mignon** reproducing player pianos equivalent to the 1924 models.

Comments:

The Welte is very sensitive to the tiniest pinholes and changes in tempo, in regard to its critical expression. Large changes in humidity and temperature and dust settling, especially tobacco smoke might also affect it, so it’s important to remember that being a pneumatic instrument which gets 100% of its expression from the timed actuation and release of the expression pneumatic, anything that can make a small difference in those factors makes a large difference in the dynamics.

In the *Original* Welte Mignon, the lock valves also utilize bleed holes to atmosphere under felt covers. These are fixed and are open to the pneumatic they control all the time, so a heavy cooking dust or smoke tar coating on the felt cover can theoretically change overall bleed characteristics, too. Vulnerable to commercial usage.

Also in some *Original* Mignons is a 4 hole air tracker that operates on the same principle as the Standard air tracking system, only it is not adjustable, except for the distance the cam is clamped from the right roll spindle. Sometimes merely adjusting the system to begin its tracking in the center of that cam when the roll is centered is all it takes to cure a tracking problem. Bench test the top action for reliable tracking with multiple rolls before placing it into the piano.

The valve bleeds in the Mignon are #70s, to make them as sensitive as possible but can also cause valve over-sensitivity in some cases if everything isn’t perfect, and that would include a slightly loose elbow or nipple, which in other pianos would be ignored. Small bleeds can make problems. You might get spurious expressions or a wrong note when pressures rise suddenly that make no sense at all. Recognize the tendency of this instrument to do that because tiny bleeds tend to clog more readily.

The overall performance of Weltes is really astoundingly realistic, and its ability to play softly equal to the Ampico B. Be exacting in regulation because it’s worth it.